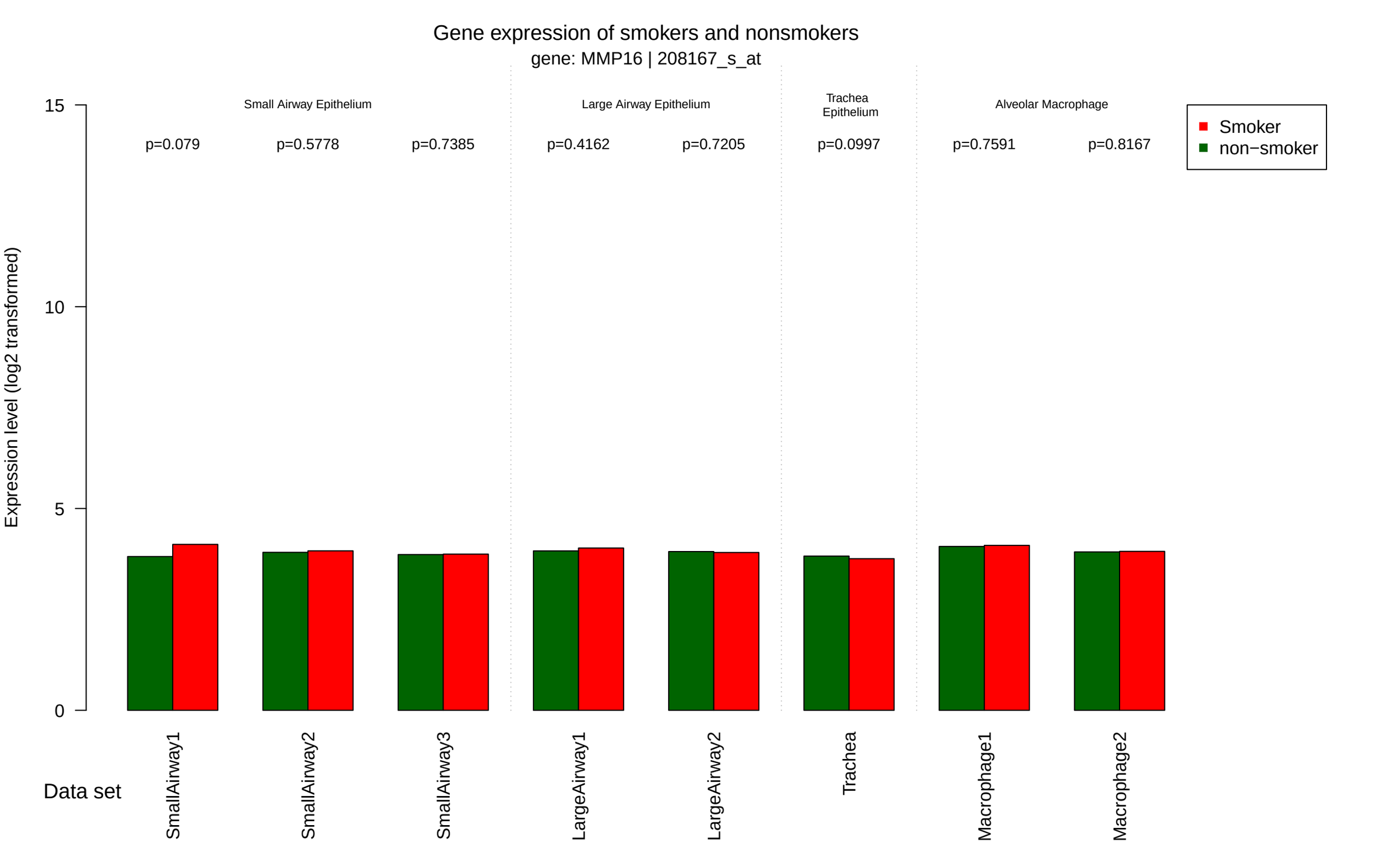
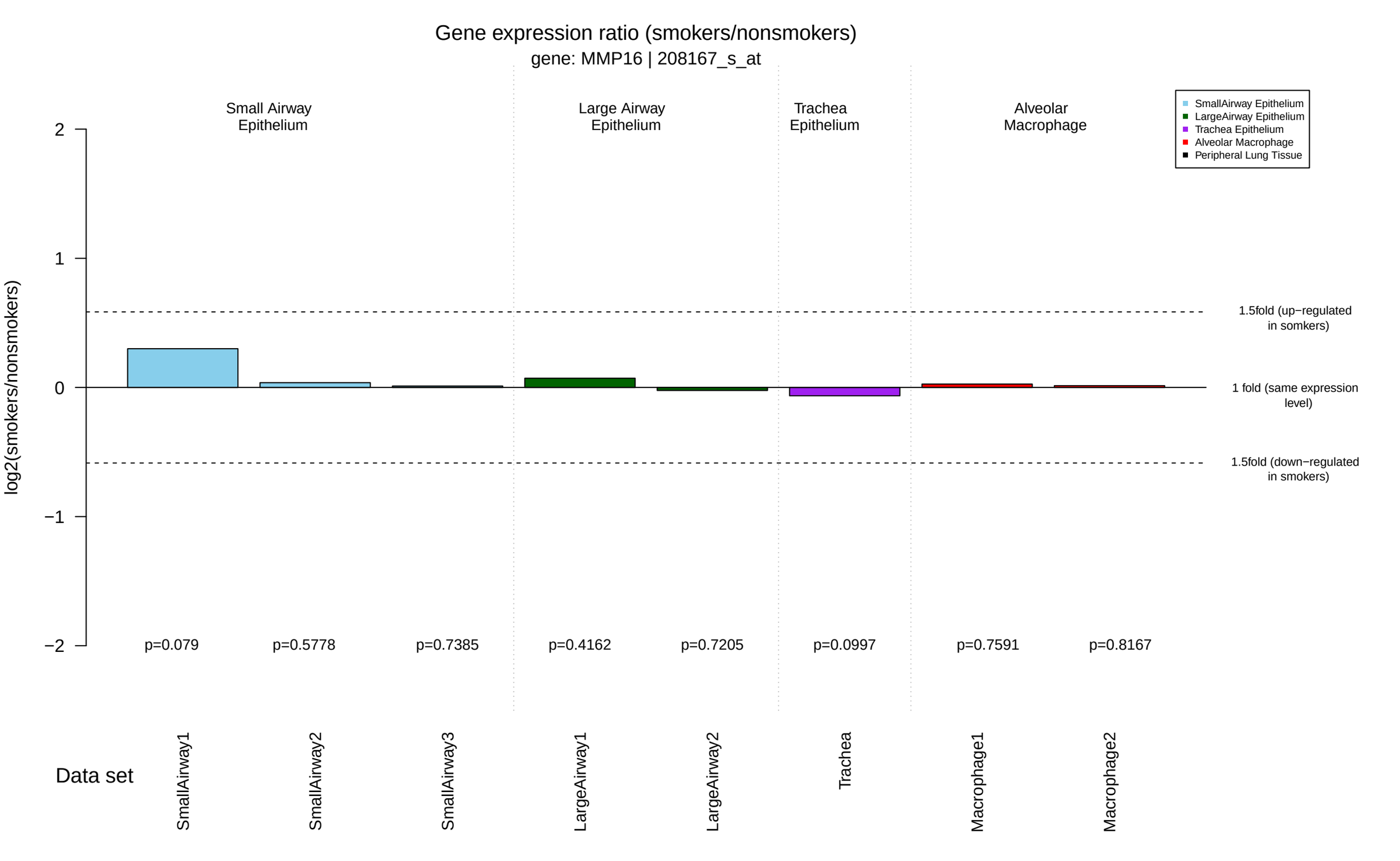
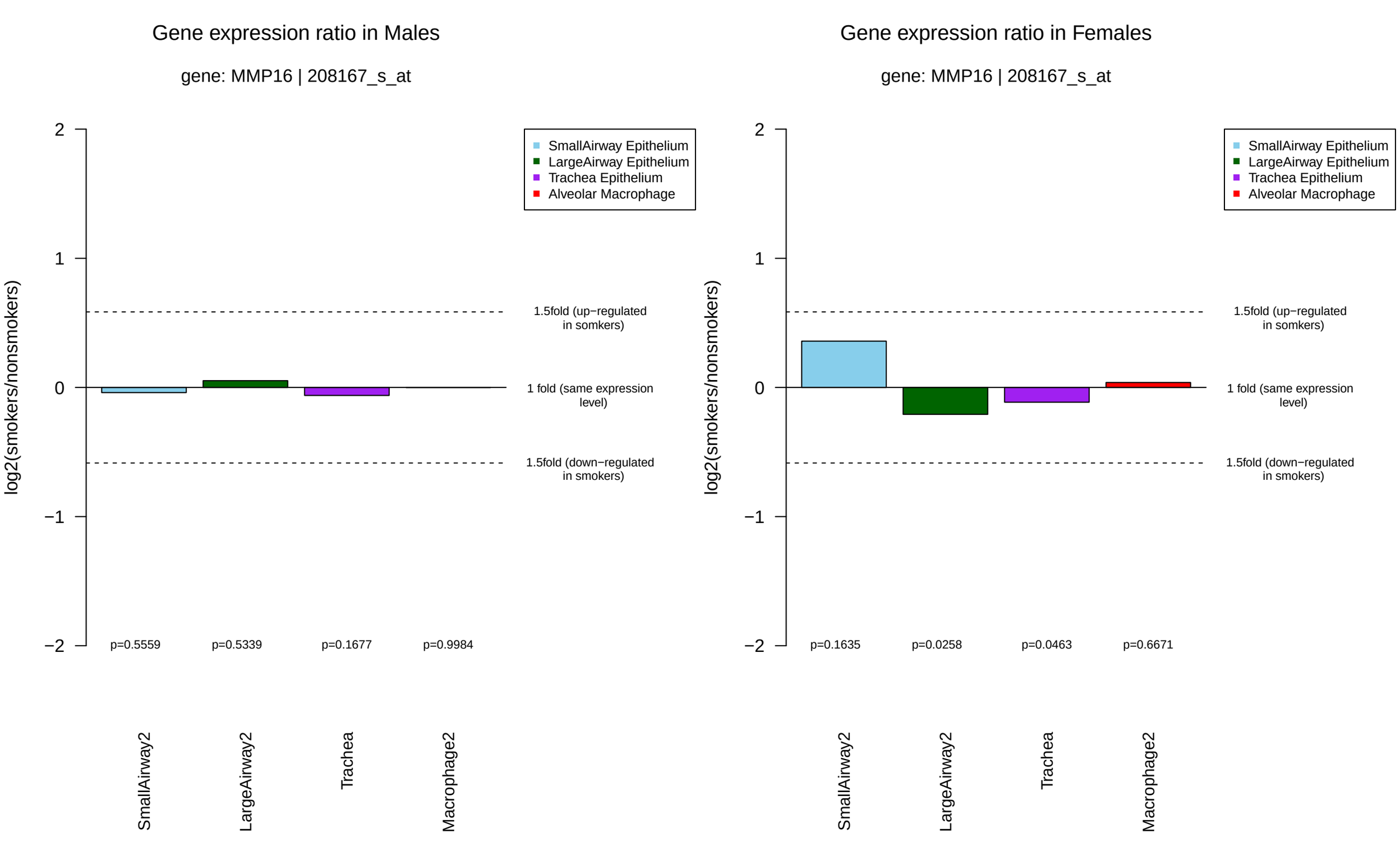
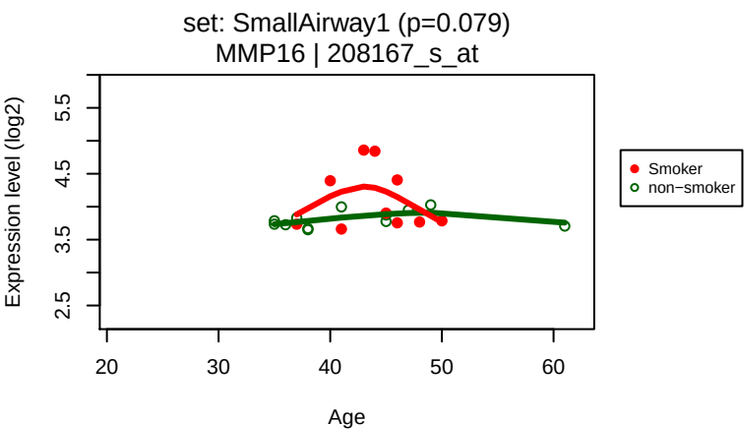
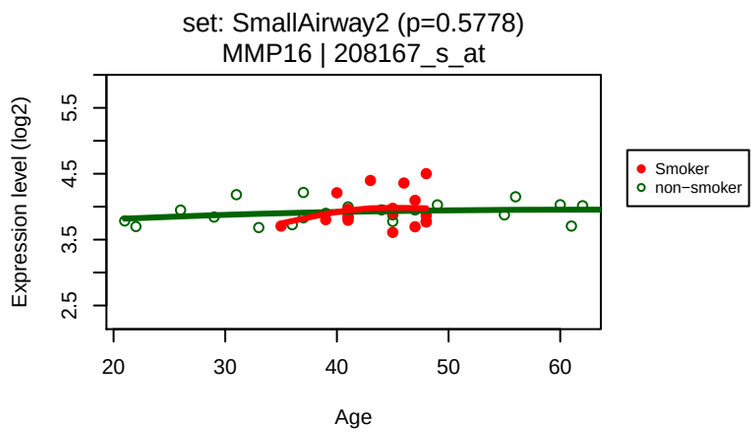
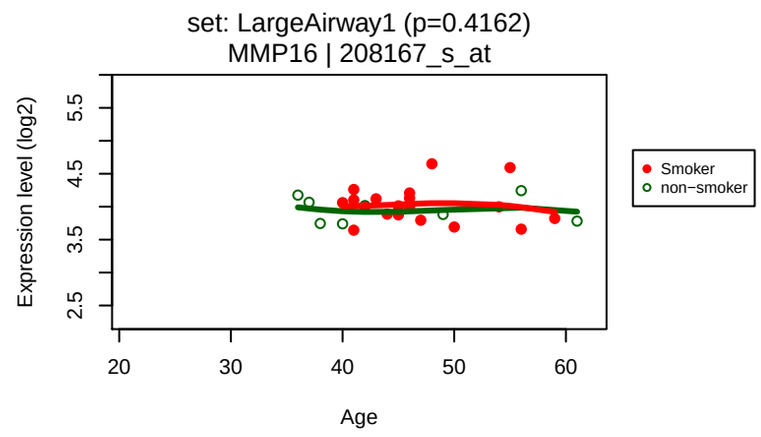

Gene expression of smokers and nonsmokers
gene: MMP16 | 208167_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.079
p=0.5778
p=0.7385
p=0.4162
p=0.7205
p=0.0997
p=0.7591
p=0.8167
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

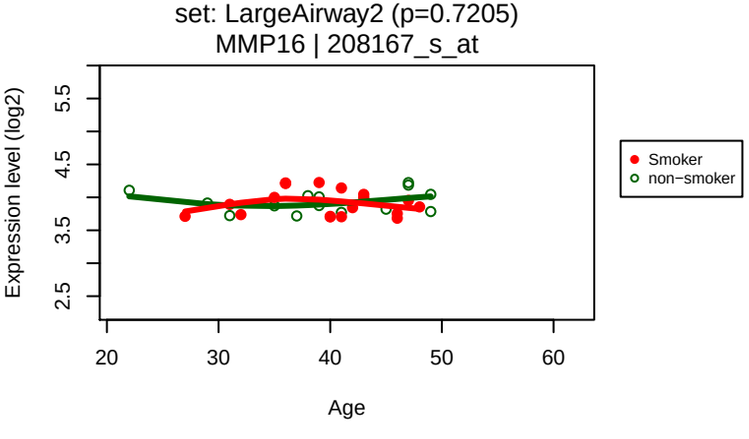

Gene expression ratio (smokers/nonsmokers)
gene: MMP16 | 208167_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.079
p=0.5778
p=0.7385
p=0.4162
p=0.7205
p=0.0997
p=0.7591
p=0.8167
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

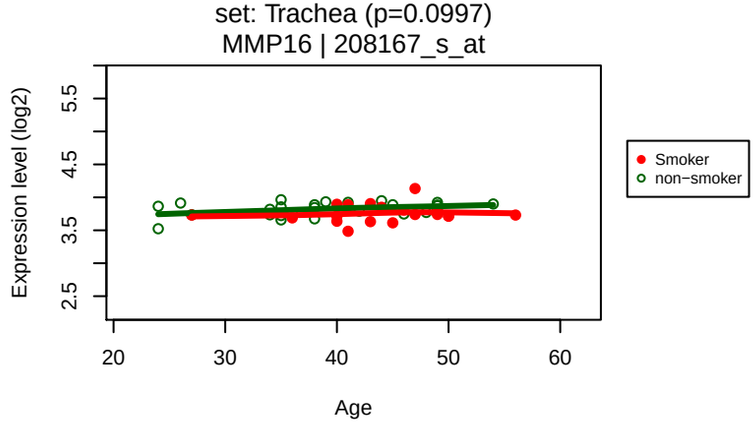

Gene expression ratio in Males
gene: MMP16 | 208167_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5559
p=0.5339
p=0.1677
p=0.9984
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP16 | 208167_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1635
p=0.0258
p=0.0463
p=0.6671
SmallAirway2
LargeAirway2
Trachea
Macrophage2

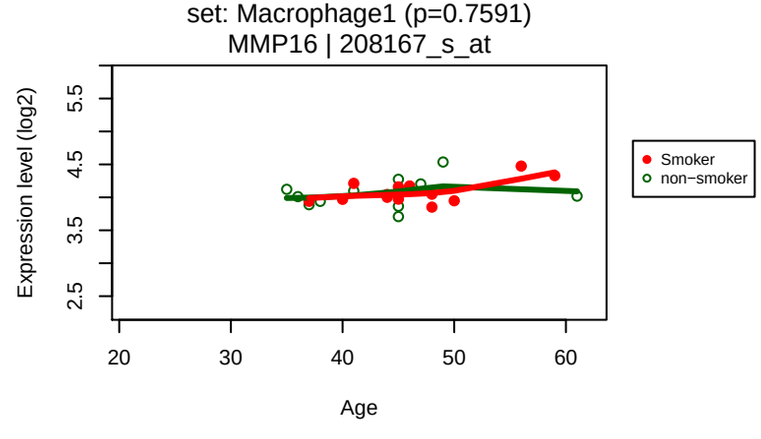

set: SmallAirway1 (p=0.079)
MMP16 | 208167_s_at
Expression level (log2)
Age
Smoker
non−smoker

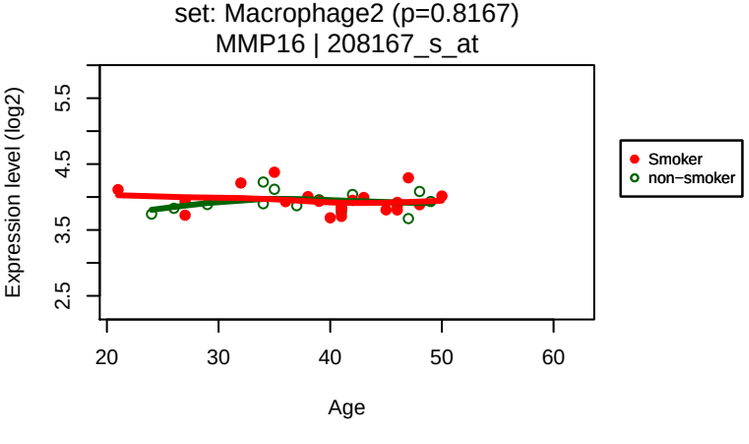

set: SmallAirway2 (p=0.5778)
MMP16 | 208167_s_at
Expression level (log2)
Age
Smoker
non−smoker

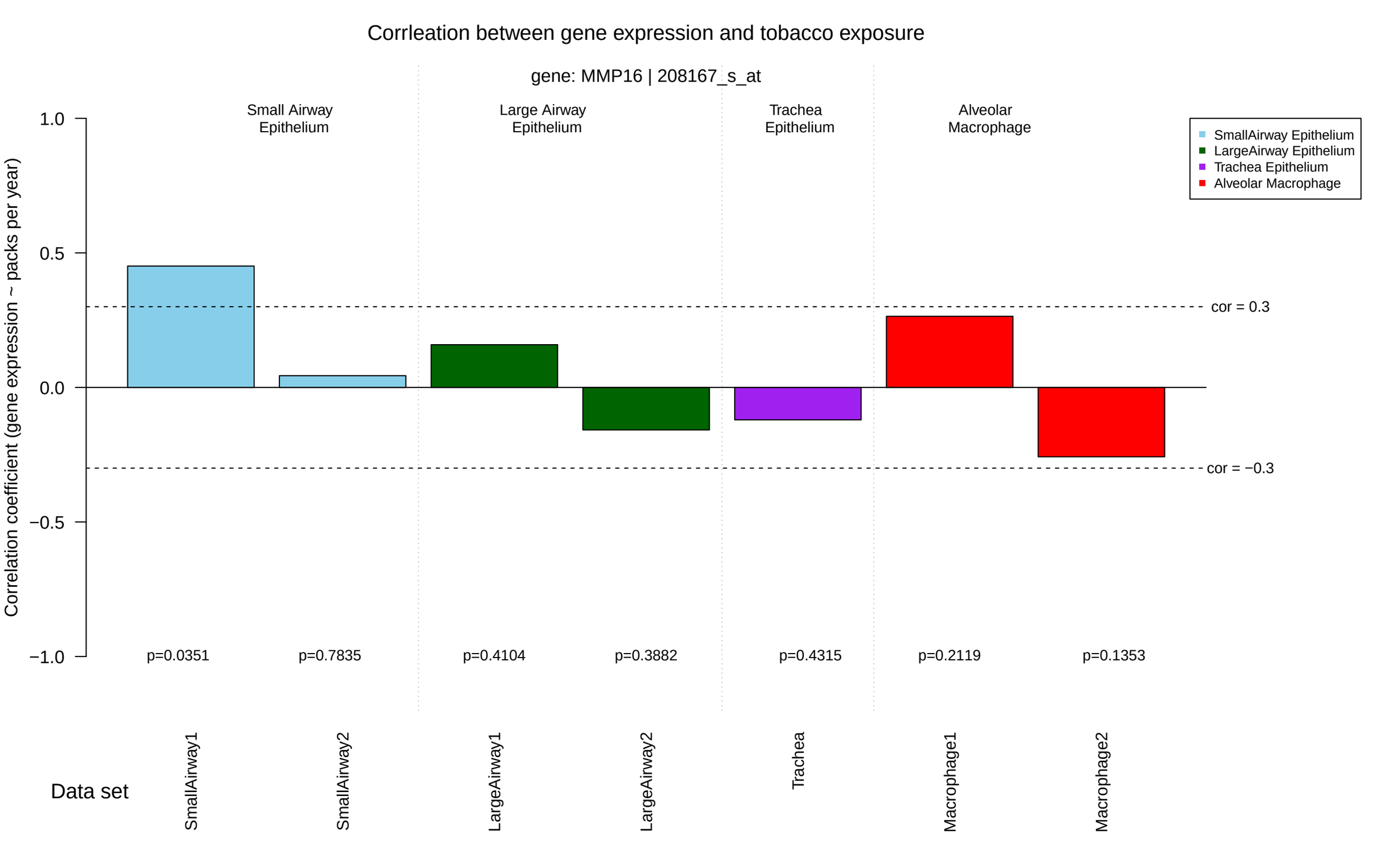

set: LargeAirway1 (p=0.4162)
MMP16 | 208167_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7205)
MMP16 | 208167_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0997)
MMP16 | 208167_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7591)
MMP16 | 208167_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8167)
MMP16 | 208167_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP16 | 208167_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0351
p=0.7835
p=0.4104
p=0.3882
p=0.4315
p=0.2119
p=0.1353
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2